The Job Sequencing and Tool Switching Problem

Part V: JGP + GSP Heuristic — Gap Analysis and Bounds

Scope and status. This note studies the additive gap $\mathrm{gap}(I) = H(I) - Z^*_{\mathrm{SSP}}(I)$ and the ratio $H(I)/Z^*_{\mathrm{SSP}}(I)$ between the cost H of the two-phase *JGP + GSP* heuristic and the optimal uniform-SSP cost Z^*_{SSP}. What is **proved** here: (i) two lower bounds on Z^*_{SSP} — the grouping bound $Z^*_{\mathrm{SSP}} \geq K^* - 1$ and the tool-counting bound $Z^*_{\mathrm{SSP}} \geq \left|\bigcup_j T_j\right| - b$ (theorems 2.1 and 2.2), together with the classical $Z^*_{\mathrm{SSP}} \leq b(K^* - 1)$; (ii) a computationally certified counterexample (the 6-job ring) with gap $= 1$ and ratio $4/3$ (example 3.1); (iii) a construction with gap $= g$ for every g, so the gap is *unbounded* (theorem 4.1); (iv) a directly computable gap bound gap $\leq H - \max(K^* - 1, \left|\bigcup_j T_j\right| - b)$ (proposition 5.1) and two sufficient conditions for zero gap (theorems 6.3 and 6.4); (v) a *grouping-exactness* theorem (theorem 6.1): Z^*_{SSP} equals the minimum GTSP cost over *all* feasible groupings, so the gap is exactly the price of using the fewest groups, and a sub-optimal grouping can strictly beat the heuristic (theorem 6.6). What is **conjectural**: the tight ratio bound $H/Z^*_{\mathrm{SSP}} \leq 4/3$ for $b = 3$ (conjecture 5.4); it is *not* proved, and the previously attempted route via $\sum_k R_k \leq H/4$ is false and is retracted (section 5.2). All numerical assertions are verified by the scripts in `plans-genai/_verification/` (see `VERIFIED_FACTS.md`); reproduce with `python3 run_study.py`.

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The JGP + GSP Heuristic

We work throughout with the *uniform* SSP: a single magazine of capacity b, jobs $j \in J$ with required tool sets $T_j \subseteq T$, $|T_j| \le b$, and switch cost equal to the number of tool insertions along the processed sequence. We adopt the *free-initial-load* convention: the first $\le b$ tools are loaded for free (equivalently, the sequence starts from a free dummy depot), so Z^*_{SSP} counts only the insertions made between consecutive jobs. For a fixed sequence the optimal loading is computed by KTNS [Tang and Denardo(1988)].

The Job Grouping Problem (JGP)

Definition 1.1 (Feasible group)**.** A set of jobs $G \subseteq J$ is *feasible* if all its tools fit simultaneously: $\left|\bigcup_{j\in G} T_j\right| \le b$.

Definition 1.2 (JGP as set covering/partitioning [Crama and Oerlemans(1994), Catanzaro et al.(2015)Catanza
Let $\mathcal{S}$ be the family of feasible groups and $y_G \in \{0,1\}$. The JGP is

$$Z^*_{\mathrm{JGP}} = \min\Big\{\textstyle\sum_{G\in\mathcal{S}} y_G : \ \sum_{G\ni j} y_G \ge 1 \ \ \forall j \in J,\ y_G \in \{0,1\}\Big\},$$

the minimum number of feasible groups ("batches") that cover J. We write $K^* = Z^*_{\mathrm{JGP}}$.

Remark 1.3 (Hardness and exact solution)**.** The JGP is NP-hard [Crama and Oerlemans(1994)]. It is solved exactly by the ARF MILP of Catanzaro et al. [Catanzaro et al.(2015)Catanzaro, Gouveia, and Labbé] and by Branch-and-Price [Otiai(2024)]; the pricing subproblem generates maximal feasible groups (a set-union knapsack).

> **Feasibility is a group property, not a pairwise one.** A group is feasible iff the union of *all* its tool sets fits in b. This is stronger than pairwise compatibility: the three jobs $\{t_1,t_2\},\{t_1,t_3\},\{t_1,t_4\}$ ($b=3$) are pairwise compatible (each pairwise union has size 3) yet jointly infeasible ($|\{t_1,t_2,t_3,t_4\}| = 4 > b$). Consequently K^* is *not* in general the chromatic number of the pairwise conflict graph; see section 8.

From Groups to Configurations and the GSP

Given a grouping $G_1,\dots,G_K$, each group G_k must be served by a *configuration* C_k (a magazine state with $|C_k| = b$) that covers it.

Definition 1.4 (Configuration cluster and slack)**.** For a group G with required set $T^{(G)} = \bigcup_{j\in G} T_j$,

$$\mathcal{C}(G) = \big\{C \subseteq T : \ |C| = b,\ T^{(G)} \subseteq C\big\}, \qquad s(G) = b - |T^{(G)}| \ge 0.$$

The $s(G)$ free slots ("slack") may be filled with tools of *adjacent* groups to lower the transition cost.

Definition 1.5 (Group Sequencing Problem and the heuristic cost H)**.** The *GSP* chooses, for a given grouping, a permutation π of the groups and a configuration $C_k \in \mathcal{C}(G_k)$ for each k, minimising

$$H(\pi,\{C_k\}) = \sum_{i=1}^{K-1} \left|C_{\pi(i+1)} \setminus C_{\pi(i)}\right| = \sum_{i=1}^{K-1} \big(b - |C_{\pi(i)} \cap C_{\pi(i+1)}|\big).$$

This is a path **Generalised TSP** over the clusters $\{\mathcal{C}(G_k)\}$. The JGP + GSP *heuristic* (1) solves the JGP to get a K^*-grouping, then (2) solves the GSP exactly; its cost is

$$H \ = \min_{\text{JGP-optimal groupings}} \ \min_{\pi,\{C_k\}} H(\pi,\{C_k\}),$$

the minimum over all K^*-group JGP solutions of the GTSP optimum. Any such schedule is a feasible SSP solution, so $H \ge Z^*_{\mathrm{SSP}}$.

Lemma 1.6 (Per-transition lower bound)**.** *For an ordered adjacent pair G_k, G_{k+1}, every choice of configurations costs at least*

$$w^*(G_k, G_{k+1}) = \max\Big(0,\ \big|T^{(G_k)} \cup T^{(G_{k+1})}\big| - b\Big).$$

For zero-slack *groups ($s(G_k) = s(G_{k+1}) = 0$) the configurations are forced ($C_k = T^{(G_k)}$) and the transition costs exactly $b - |T^{(G_k)} \cap T^{(G_{k+1})}|$.*

Proof. $C_{k+1} \supseteq T^{(G_{k+1})}$ has $s(G_{k+1})$ free slots; of the $|T^{(G_k)} \setminus T^{(G_{k+1})}|$ tools that C_k holds but G_{k+1} does not require, at most $s(G_{k+1})$ can be retained, so at least $\max(0, |T^{(G_k)} \setminus T^{(G_{k+1})}| - s(G_{k+1})) = \max(0, |T^{(G_k)} \cup T^{(G_{k+1})}| - b)$ must leave. With zero slack no tool can be retained beyond the required set, giving the stated exact cost. □

Greedy "lookahead" is not globally optimal for the GSP. Filling each group's slack with the next group's tools (lookahead) attains the per-transition bound of lemma 1.6 for one isolated transition, but the padding of C_k influences *both* the $(k-1, k)$ and $(k, k+1)$ transitions, so summing per-transition minima can under-count. The true heuristic cost H is the GTSP optimum of definition 1.5, which we compute exactly. For *zero-slack* instances there is no padding freedom and the GSP reduces to a TSP over forced configurations; all worst-case instances in this note are zero-slack, where the per-transition formula is exact.

Remark 1.7 (Standing in the literature)**.** The two-phase decomposition studied here is the method of practice, not a straw man: in colour printing, an SSP in disguise, [Burger et al.(2015)Burger, Jacobs, va implement exactly JGP + GSP (unicost set covering for the grouping, a TSP over groups for the sequencing), enumerate optimal groupings via Stirling numbers, and measure the decomposition's suboptimality empirically, finding it small on industrial instances—consistent with earlier reports they survey. No *worst-case* analysis of the decomposition (gap or ratio bounds, extremal families, exactness conditions) appears in the literature, including the survey of [Calmels(2018)]; the approximation-flavoured results that exist for the SSP concern other heuristics. That missing analysis is what this note develops.

Lower and Upper Bounds Relating Z^*_{SSP}, K^* and the Tool Count

Two elementary but distinct lower bounds control Z^*_{SSP}. Both are used repeatedly.

Theorem 2.1 (Grouping lower bound)**.** $Z^*_{\text{SSP}} \geq K^* - 1$.

Proof. An optimal SSP solution visits a sequence of magazine states; let $W_1, \ldots, W_m$ be the distinct states in order of first appearance. Each job is served by some W_r, so $\{W_1, \ldots, W_m\}$ covers J by feasible configurations and $m \geq K^*$. Consecutive distinct states differ in ≥ 1 tool, so the number of insertions is $\geq m - 1 \geq K^* - 1$. □

Theorem 2.2 (Tool-counting lower bound)**.** *Let $U = \bigcup_{j \in J} T_j$. Then $Z^*_{\text{SSP}} \geq |U| - b$.*

Proof. Every tool of U is required by some job, hence is loaded into the magazine at least once. At most b tools occupy the initial free load; every other loading is a counted insertion. Therefore the number of insertions is at least $|U| - b$. □

Remark 2.3 (Relation to the SSPMF bound). $|U| - b$ is exactly the $M - C$ linear-programming lower bound of the multicommodity-flow formulation (SSPMF) of da Silva et al. [da Silva and Yanasse(2024)] ($M = |U|$ tools, $C = b$ capacity — the identification $M = |U|$ assumes every tool is required by some job, w.l.o.g. after deleting unused tools); theorem 2.2 gives the elementary combinatorial proof. Combining the two bounds,

$$\boxed{Z^*_{\mathrm{SSP}} \;\geq\; \max\big(K^* - 1,\; |U| - b\big).}$$

Theorem 2.4 (Grouping upper bound). $Z^*_{\mathrm{SSP}} \leq b\,(K^* - 1)$.

Proof. Process a K^*-group JGP solution batch by batch under one fixed covering configuration per batch. The initial load is free; each of the $K^* - 1$ inter-batch transitions inserts at most b tools. □

Which bound is tight? Neither lower bound is tight in general, and their maximum need not equal Z^*_{SSP}. On the ring family of section 3 and on the entire exhaustively enumerated $b = 3$ edge family (jobs = graph edges, $|T_j| = 2$; 10691 instances) we verified $Z^*_{\mathrm{SSP}} = \max(K^* - 1, |U| - b)$ with no exception. But this equality is *family-specific*: among $b = 3$ instances with $|T_j| \leq 3$ on ≤ 5 tools, 2410 of 21569 have Z^*_{SSP} strictly above *both* bounds, and the $b = 4$ sliding window $T_{j_i} = \{i, i{+}1, i{+}2\}$ mod 8 has $Z^*_{\mathrm{SSP}} = 5 > \max(3, 4)$. Only the inequality of remark 2.3 is universal.

A Certified Counterexample: the 6-Job Ring

Example 3.1 (6-job ring). Let $T = \{t_1, \dots, t_6\}$, $b = 3$, and $T_{j_i} = \{t_i, t_{i+1 \bmod 6}\}$ for $i = 1, \dots, 6$: consecutive jobs share one tool.

JGP optimum ($K^* = 3$). The minimum number of feasible groups is 3 (two optimal groupings exist). One is $G_1 = \{j_1, j_2\}$, $G_2 = \{j_3, j_4\}$, $G_3 = \{j_5, j_6\}$ with required sets $T^{(1)} = \{t_1, t_2, t_3\}$, $T^{(2)} = \{t_3, t_4, t_5\}$, $T^{(3)} = \{t_1, t_5, t_6\}$, each of size 3 (zero slack), so configurations are forced.

JGP + GSP cost ($H = 4$). With forced configurations the GSP is a TSP on three nodes; every pair of the three sets shares exactly one tool, so each transition costs $b - 1 = 2$ and any order pays $2 \times 2 = 4$. Thus $H = 4$.

SSP optimum ($Z^*_{\mathrm{SSP}} = 3$). The sequence $j_1 \to \cdots \to j_6$ under KTNS:

Step	Job	Magazine	Switches
1	j_1	$\{t_1, t_2, t_3\}$	(free initial load)
2	j_2	$\{t_1, t_2, t_3\}$	0
3	j_3	$\{t_1, t_3, t_4\}$	1 $(-t_2, +t_4)$
4	j_4	$\{t_1, t_4, t_5\}$	1 $(-t_3, +t_5)$
5	j_5	$\{t_1, t_5, t_6\}$	1 $(-t_4, +t_6)$
6	j_6	$\{t_1, t_5, t_6\}$	0

KTNS retains t_1 as a *bridge* across jobs j_2–j_5, so $Z^*_{\mathrm{SSP}} = 3$, matching both lower bounds: $|U| - b = 6 - 3 = 3$ (and $K^* - 1 = 2$).

Gap. $H - Z^*_{\mathrm{SSP}} = 4 - 3 = 1$; ratio $4/3$. Verified exactly by state-DP and by PORTA enumeration.

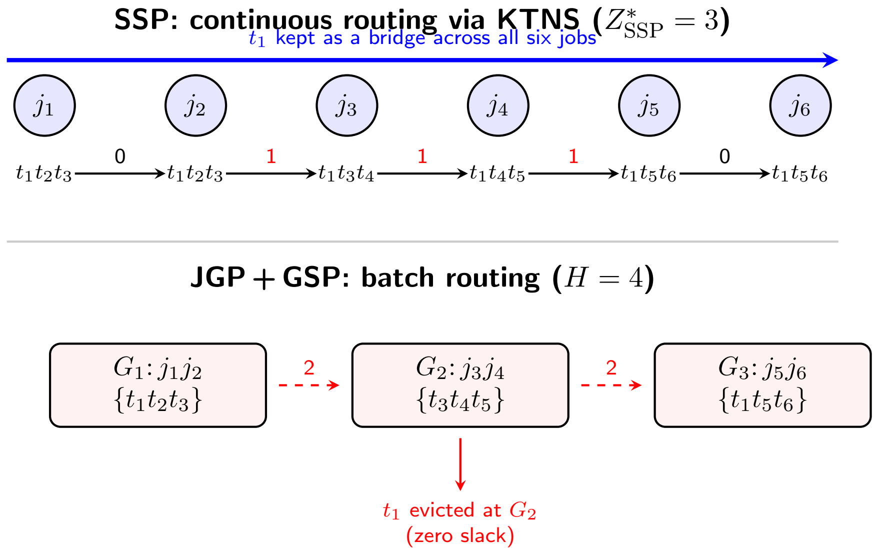

Figure 1: The 6-ring. KTNS keeps t_1 resident through all jobs (3 switches). The 3-group JGP saturates each batch with its required tools, forcing t_1 out at G_2 and back in at G_3 (4 switches). Gap $= 1$.

The Gap is Unbounded

Theorem 4.1 (Gap grows with K^*)**.** *For every integer $g \geq 1$ there is a uniform-SSP instance with $K^* = 3g$ on which the JGP + GSP heuristic costs exactly g more switches than the optimum: $Z^*_{\mathrm{SSP}} = 6g - 3$, $H = 7g - 3$, gap $= g$.*

Proof. Take g tool-disjoint copies of the 6-ring (example 3.1); copy ℓ uses tools $\{t_{6\ell-5}, \ldots, t_{6\ell}\}$, so $|U| = 6g$.

Optimum. By theorem 2.2, $Z^*_{\mathrm{SSP}} \geq |U| - b = 6g - 3$. Processing the copies contiguously achieves this value (within a copy, 3 switches as in example 3.1; between two copies the magazine swaps completely, costing $b = 3$, of which the first such load per copy realises that copy's tool count); the explicit schedule attains $6g - 3$, so $Z^*_{\mathrm{SSP}} = 6g - 3$.

Heuristic. The clusters of different copies are tool-disjoint, so every inter-copy transition costs $b = 3$ and the GTSP decomposes: $H = \sum_\ell H^{(\ell)} + b(g-1) = 4g + 3(g-1) = 7g - 3$.

Hence gap $= H - Z^*_{\mathrm{SSP}} = (7g - 3) - (6g - 3) = g$. $\square$

Remark 4.2 (The gap scales with K^*, not b)**.** Here gap $= g = K^*/3$, consistent with the worst-case envelope gap $\leq H \leq b(K^* - 1)$ from theorem 2.4. The relevant complexity parameter for the additive gap is the number of groups K^*, not the capacity b. The ratio, by contrast, *shrinks* as g grows: $(7g-3)/(6g-3)$ decreases from $4/3$ at $g = 1$ to $7/6$ as $g \to \infty$ (section 5). All values verified for $g = 1, \ldots, 4$.

Remark 4.3 (Two informal "sources" of gap)**.** It is tempting to split the gap into a *padding* part (even the best configurations for a fixed K^*-grouping cost more than Z^*_{SSP}) and a *grouping* part (the SSP optimum need not respect any K^*-grouping). This is a useful intuition but not a theorem: the two effects are not independent and there is no proved decomposition. In the 6-ring, exhaustive enumeration of all 18 feasible groupings shows the minimum GTSP cost over K^*-groupings is 4 while over *all* groupings it is 3; thus for this instance the entire heuristic gap is removed by changing the grouping, not the padding — see theorem 6.6.

A Computable Gap Bound, the Bridge Mechanism, and the Ratio

A directly computable gap bound

Proposition 5.1 (Gap bound)**.** *For every instance,*

$$0 \;\le\; \mathrm{gap}(I) = H - Z^*_{\mathrm{SSP}} \;\le\; H - \max\bigl(K^* - 1,\; |U| - b\bigr).$$

Proof. $H \ge Z^*_{\mathrm{SSP}}$ because the heuristic schedule is SSP-feasible; the upper bound substitutes the proved lower bound $Z^*_{\mathrm{SSP}} \ge \max(K^* - 1, |U| - b)$ (remark 2.3). □

For the 6-ring this gives gap $\le 4 - \max(2, 3) = 1$, which is tight. The bound is computable without solving the SSP, since H, K^* and $|U|$ are all available from the heuristic.

The bridge / re-entry mechanism

The gap is caused by tools the SSP keeps resident across a group boundary but the heuristic must evict and reload.

Definition 5.2 (Bridge tool)**.** Fix a K^*-grouping in GSP-optimal order. A tool t is a *bridge* if it is required in some group G_k and again in some later $G_{k'}$ ($k' > k$) but not required by an intermediate group whose forced configuration must therefore evict it. KTNS can keep t resident through the intermediate jobs using slack, saving one insertion that the heuristic pays.

In the 6-ring, t_1 is required by $j_1 \in G_1$ and $j_6 \in G_3$ but not by $G_2 = \{j_3, j_4\}$ (whose forced configuration is $\{t_3, t_4, t_5\}$). KTNS keeps t_1 through $j_2, \ldots, j_5$; the heuristic evicts it at the $G_1 \to G_2$ boundary and reloads it at $G_2 \to G_3$, which is the one extra switch.

Retraction. An earlier draft claimed an intermediate inequality $\sum_k R_k \le H/4$ (with R_k the number of re-entering tools at boundary k) and chained it to $H/Z^*_{\mathrm{SSP}} \le 4/3$. The intermediate inequality is **false**: the zero-slack triple $T^{(1)} = \{a, b, c\}$, $T^{(2)} = \{a, b, d\}$, $T^{(3)} = \{c, d, e\}$ ($b = 3$) in GSP order has $H = 3$ and $\sum_k R_k = 1 > H/4 = 0.75$ (verified). Hence *there is no proof of the 4/3 ratio along this route*; the ratio bound below is stated as a conjecture.

The approximation ratio for $b = 3$

Proposition 5.3 (Ratio lower bound)**.** $\displaystyle\sup_{I:\, b=3} \frac{H(I)}{Z^*_{\mathrm{SSP}}(I)} \;\ge\; \frac{4}{3}$, *witnessed by the 6-ring ($H = 4$, $Z^*_{\mathrm{SSP}} = 3$).*

Conjecture 5.4 (Tight ratio for $b = 3$)**.** *For all uniform-SSP instances with* $b = 3$, $\dfrac{H}{Z^*_{\mathrm{SSP}}} \le \dfrac{4}{3}$, *with equality only at the* 6*-ring (up to relabelling).*

Status of conjecture 5.4: open, strong computational support.

- Exhaustive search over the $b = 3$ edge family (jobs = edges of a graph on ≤ 6 tool-vertices, ≤ 6 edges; 10691 instances): the maximum ratio observed is exactly $4/3$ and is *never* exceeded; every positive-gap instance there has $K^* = 3$, $|U| = 6$, $Z^*_{\mathrm{SSP}} = 3$, $H = 4$.
- The unbounded-gap family (theorem 4.1) has ratios in $[\,7/6,\ 4/3\,]$, supremum $4/3$ at $g = 1$.
- Rings of length $k = 6, 7, 8$ give ratios $4/3$, $5/4$, $6/5$; shorter rings have gap 0.

No $b = 3$ instance with ratio $> 4/3$ is known. A proof (or refutation) is the main open problem of this note; section 7 gives a general-b programme (unconditional bound theorem 7.3, conjecture conjecture 7.6, and a conditional route to $4/3$, proposition 7.16). For $b \ge 4$ the extremal ratio is *not* bounded by $4/3$: a broadened search reaches (exactly) $4/3$ at $b = 4$ and $3/2$ at $b = 5$, so $\sup_I H/Z^*_{\mathrm{SSP}}$ appears to grow with b — the $4/3$ bound is a $b = 3$ phenomenon.

When is the Gap Zero?

Since $H \ge Z^*_{\mathrm{SSP}} \ge \max(K^* - 1, |U| - b)$, the heuristic is optimal whenever it meets either lower bound. The next theorem, however, is the real organising principle: it shows the gap is *exactly* the cost of the heuristic's insistence on the minimum number of groups.

Grouping exactness: the gap is the price of minimum K^*

For a grouping $\mathcal{G}$ (any partition of J into feasible groups, of any cardinality), write $\mathrm{GTSP}(\mathcal{G})$ for its path-GTSP optimum (definition 1.5).

Theorem 6.1 (Grouping exactness)**.** *For every uniform-SSP instance,*

$$Z^*_{\mathrm{SSP}} \;=\; \min_{\mathcal{G}\ \text{feasible}} \ \mathrm{GTSP}(\mathcal{G}),$$

the minimum being over all *feasible groupings. Consequently the JGP + GSP gap is*

$$H - Z^*_{\mathrm{SSP}} \;=\; \underbrace{\min_{|\mathcal{G}|=K^*} \mathrm{GTSP}(\mathcal{G})}_{=H} \;-\; \underbrace{\min_{\mathcal{G}} \mathrm{GTSP}(\mathcal{G})}_{=Z^*_{\mathrm{SSP}}},$$

*i.e. precisely the penalty for restricting the heuristic to JGP-*optimal *(minimum-cardinality) groupings. With unrestricted grouping choice the framework is exact.*

Proof. ($\le$) Any feasible grouping together with a GSP ordering and one configuration per group is a feasible SSP schedule of cost $\mathrm{GTSP}(\mathcal{G})$, so $Z^*_{\mathrm{SSP}} \le \min_{\mathcal{G}} \mathrm{GTSP}(\mathcal{G})$. ($\ge$) Let σ be an optimal SSP schedule with full-magazine trajectory $M_1, \ldots, M_n$ (one state per job; full w.l.o.g.). Bring it into *normal form*: collapse consecutive equal states; if a configuration recurs at a later, non-consecutive visit, reassign the jobs served there to its first visit (the configuration is identical) and delete the later visit, joining its neighbours $X \to C \to Y$ into $X \to Y$ — by the triangle inequality $|Y \setminus X| \le |C \setminus X| + |Y \setminus C|$ the cost does not increase; delete any state serving no job by the same shortcut. The result is a sequence of pairwise-distinct configurations $C_{(1)}, \dots, C_{(m)}$, each serving at least one job; it is itself a feasible schedule of cost $\le Z^*_{\mathrm{SSP}}$, hence exactly Z^*_{SSP} by optimality. Assign each job to the configuration serving it: the jobs of $C_{(i)}$ have tool union contained in $C_{(i)}$, hence form a feasible group, and these groups partition J. The configurations $C_{(1)}, \dots, C_{(m)}$ in order constitute a GTSP path over this grouping of cost exactly Z^*_{SSP}, so $\min_{\mathcal{G}} \mathrm{GTSP}(\mathcal{G}) \le Z^*_{\mathrm{SSP}}$. Equality follows. (Confirmed on 1306 instances with no exception; `plans-genai/_verification/study_op.py`.) □

Remark 6.2 (Interpretation)**.** theorem 6.1 relocates the entire difficulty of the SSP into a single combinatorial choice: *which* grouping to use. The JGP solves for the *fewest* groups; the SSP wants the *cheapest-to-sequence* groups, which may be more numerous. The bridge tools of definition 5.2 are exactly the tools that a finer grouping can keep resident. This is the precise sense in which "the gap is the price of minimum K^*", and it motivates optimising over groupings directly (Part VI).

The heuristic meets a lower bound

Theorem 6.3 (Sufficient condition A — grouping bound met)**.** *If some JGP-optimal grouping admits a group order whose GTSP cost equals $K^* - 1$, then* gap $= 0$ *and* $Z^*_{\mathrm{SSP}} = H = K^* - 1$.

Proof. $H \le K^* - 1 \le Z^*_{\mathrm{SSP}} \le H$ by theorem 2.1 and $H \ge Z^*_{\mathrm{SSP}}$; all are equal. □

Theorem 6.4 (Sufficient condition B — tool bound met)**.** *If some JGP-optimal grouping has GTSP cost equal to $|U| - b$, then* gap $= 0$ *and* $Z^*_{\mathrm{SSP}} = H = |U| - b$.

Proof. $H \le |U| - b \le Z^*_{\mathrm{SSP}} \le H$ by theorem 2.2 and $H \ge Z^*_{\mathrm{SSP}}$. □

Remark 6.5 (Verified picture for $b = 3$)**.** Across the 10691-instance edge family, every zero-gap instance meets one of the two bounds (the heuristic GTSP cost equals $\max(K^* - 1, |U| - b)$), and every positive-gap instance has $K^* = 3$. So $K^* \le 2$ already forces gap $= 0$ in this family (now a theorem for all instances and all b: corollary 7.4). A full structural characterisation of the zero-gap set (a necessary-and-sufficient condition) is open.

Sub-optimal groupings can beat the JGP-optimal heuristic

The heuristic insists on a K^*-grouping. By theorem 6.1 some grouping always attains Z^*_{SSP}, and when no K^*-grouping does, a larger one must — relaxing $|\mathcal{G}| = K^*$ strictly helps. The following is the smallest explicit witness.

Theorem 6.6 (A sub-optimal grouping closes the ring gap)**.** *For the* 6*-ring, the* $K = 4$ *grouping* $\{j_1, j_2\}, \{j_3, j_4\}, \{j_5\}, \{j_6\}$ *admits configurations and an order with total GTSP cost* $3 = Z^*_{\mathrm{SSP}}$*. Hence a grouping with $K > K^*$ achieves cost strictly below the JGP-optimal heuristic value $H = 4$.*

Proof. The singletons $\{j_5\}$ and $\{j_6\}$ have slack 1, so their configurations can retain a non-required "bridge" tool. Order the four groups and pad as

$$\underbrace{\{t_1, t_2, t_3\}}_{\{j_1, j_2\}} \to \underbrace{\{t_1, t_3, t_6\}}_{\{j_6\}} \to \underbrace{\{t_3, t_5, t_6\}}_{\{j_5\}} \to \underbrace{\{t_3, t_4, t_5\}}_{\{j_3, j_4\}}.$$

Each configuration covers its group ($\{j_6\}$ needs $\{t_1, t_6\}$, $\{j_5\}$ needs $\{t_5, t_6\}$), and consecutive configurations share two tools, so every transition costs 1: total $3 = Z^*_{\text{SSP}}$ (the tool t_3 now plays the bridge role across $\{j_6\}, \{j_5\}$). Exhaustive evaluation of all 18 feasible groupings (script `run_study.py`) confirms the minimum GTSP cost over $K = 4$ groupings is 3, versus 4 over all $K^* = 3$ groupings. □

Remark 6.7 (Consequence)**.** This answers, affirmatively, the question of whether sub-optimal JGP solutions can lower the SSP cost of the decomposition: *yes.* It also refutes the plausible-looking claim that "extra groups can only add transitions" for zero-slack instances — splitting a zero-slack group creates slack in the parts, which can enable bridging the heuristic otherwise loses. The right object to optimise is therefore not Z^*_{JGP} but the joint (grouping, GSP) cost; this is the subject of Part VI (`06_grouping_selection.tex`).

General-b Bounds on the Ratio

Throughout this section $m = |U|$, $q = m - b$, and $K^* \geq 2$ (the case $K^* = 1$ is trivial, $H = Z^*_{\text{SSP}} = 0$). Recall $Z^*_{\text{SSP}} \geq \max(K^* - 1, q)$ (remark 2.3).

The cost identity and re-insertions

Lemma 7.1 (Cost identity)**.** *W.l.o.g. every magazine state of a full-magazine schedule (and every GSP configuration) is a subset of U (pad within U; possible since $m \geq b$ when $K^* \geq 2$). Then the cost of any such schedule equals*

$$\text{cost} \;=\; (m - b) \;+\; R,$$

where $R \geq 0$ counts the re-insertions*: insertions of tools previously present and evicted (including evicted tools of the free initial load).*

Proof. The initial load holds b tools of U for free. Each of the remaining $m - b$ tools of U is required by some job, hence inserted at least once. Splitting the total insertions into first-time insertions (exactly $m - b$) and all others (by definition R) gives the identity. □

Consequently $Z^*_{\text{SSP}} = q + R_{\text{opt}}$ and $H = q + R_H$, so gap $= R_H - R_{\text{opt}}$: *the gap is exactly the excess re-insertions forced by the K^*-grouping.* This makes the bridge mechanism of definition 5.2 quantitative.

Unconditional bounds

Lemma 7.2 (Transition cap)**.** *In any GSP path with configurations $\subseteq U$, every transition costs $|C_{k+1} \setminus C_k| \leq \min(b, q)$. Hence $H \leq (K^* - 1) \cdot \min(b, q)$.*

Proof. $|C_{k+1} \setminus C_k| \le |C_{k+1}| = b$ and $|C_{k+1} \setminus C_k| \le |U \setminus C_k| = m - b = q$ since $C_{k+1} \subseteq U$ and $|C_k| = b$. A covering configuration within U exists for every group, so the heuristic may be restricted to such configurations w.l.o.g. □

Theorem 7.3 (Unconditional ratio bound)**.** *For every uniform-SSP instance with $K^* \ge 2$,*

$$\frac{H}{Z^*_{\mathrm{SSP}}} \;\le\; \frac{(K^*-1)\min(b,q)}{\max(K^*-1,\,q)} \;\le\; \min\bigl(b,\;K^*-1,\;q\bigr).$$

Proof. $Z^*_{\mathrm{SSP}} \ge \max(K^*-1, q) \ge 1$, so divide lemma 7.2 by it. For the second inequality: $(K^*-1)\,q/\max(K^*-1,q) = \min(K^*-1,q)$ exactly, and $\min(b,q) \le q$ gives the $\min(K^*-1,q)$ part, while $(K^*-1)/\max(K^*-1,q) \le 1$ and $\min(b,q) \le b$ give the b part. □

Corollary 7.4 (Zero-gap classes, all b)**.** *(i) If $K^* = 2$ then* gap $= 0$ *(the ratio bound is $\le K^*-1 = 1$); in fact $H = Z^*_{\mathrm{SSP}} = m - b$, attained by padding C_1 with tools of $T^{(2)}$ and C_2 with tools of C_1. (ii) If $m \le b+1$ then* gap $= 0$*. (iii) If $K^* = 3$ then $H/Z^*_{\mathrm{SSP}} \le 2$.*

Proof. All three are instances of theorem 7.3 ($K^*-1 = 1$; $q \le 1$; $K^*-1 = 2$). For the explicit construction in (i): $C_1 = T^{(1)} \cup P$ with $P \subseteq T^{(2)} \setminus T^{(1)}$, $|P| = b - |T^{(1)}|$ (possible since $m \ge b$), and $C_2 = T^{(2)} \cup (\text{pad from } C_1)$; then $|C_2 \setminus C_1| = |T^{(2)} \setminus C_1| = (m - |T^{(1)}|) - (b - |T^{(1)}|) = m - b \le Z^*_{\mathrm{SSP}} \le H$. □

Remark 7.5. corollary 7.4(i) upgrades the empirical observation of section 6 ("$K^* \le 2$ forces gap $= 0$ in the $b = 3$ edge family") to a theorem for *all* instances and *all* b. Worst cases therefore require $K^* \ge 3$, consistent with every positive-gap instance found so far having $K^* \in \{3, 4\}$.

The excess re-insertion conjecture

All known extremal instances satisfy a much stronger bound than theorem 7.3, with *equality*:

Conjecture 7.6 (Excess re-insertions; generalises the K^*=3, b=3 conjecture)**.** *For every uniform-SSP instance with $K^* \ge 2$,*

$$\text{gap} \;=\; H - Z^*_{\mathrm{SSP}} \;\le\; K^* - 2.$$

REFUTED (2026-07-02). conjecture 7.6 is **false**. Counterexample ($b = 5$, $U = \{0, \ldots, 8\}$, $K^* = 3$): $T = (\{0,1,2,4,7\}, \{0,4,5\}, \{1,5\}, \{2,6,8\}, \{3,4,5,8\})$ has $q = 4$, $Z^*_{\mathrm{SSP}} = 4$, $H = 6$: gap $2 > K^* - 2 = 1$, ratio $3/2$. A second witness with the same profile: $T = (\{0,2,3\}, \{0,6,7,8\}, \{1,2,4,6\}, \{1,3\}, \{2,5,7\}, \{4,8\})$. Both verified exhaustively by two independent implementations (`_verification/verify_gapK_refutation.py`). They sit exactly on the boundary $2b = q + 3R + 6$ of the proved regimes and attain theorem 7.12's bound with equality. The earlier evidence ($\sim$1380 instances, $b \in \{2, 3, 4\}$) covered precisely the region where the bound is a *theorem*; $b = 5$ had never been tested. Consequences: proposition 7.15 has a false hypothesis and is void; the quantitative bounds theorem 7.12 and theorem 7.14 are the standing worst-case results, tight at every known extremal instance;

ratio $3/2$ occurs already at $K^* = 3$, so the $2 - 2/K^*$ extremal pattern is dead; the open question is the exact extremal law.

Evidence for conjecture 7.6. $K^* = 2$ is *proved* (corollary 7.4). Every known extremal witness attains the bound with equality: the 6-ring (K^*=3, gap=1), the b=4 witness (K^*=3, gap=1), the b=5 witness (K^*=4, gap=2); the g-copy family has gap $= g \le 3g - 2$. No violation in ~1380 instances ($b \in \{2, 3, 4\}$, mixed job sizes; `_verification/verify_ratio_section.py`), nor anywhere in the earlier exhaustive searches. The $K^* = 3$ case is *proved* for every $b \le 4$ and, in general, outside an explicit corner (section 7.4, theorem 7.12); **in that corner the conjecture fails** — see the refutation box above.

Partial resolution of conjecture 7.6: the $K^* = 3$ case

Throughout, configurations are w.l.o.g. subsets of U and $R = Z^*_{\mathrm{SSP}} - q \ge 0$ denotes the optimal re-insertion count (lemma 7.1).

Lemma 7.7 (Path re-insertion formula)**.** *Let $C_1, \dots, C_p \subseteq U$ be a configuration path ($|C_l| = b$, consecutive entries distinct, $\bigcup_l C_l = U$). Its free-initial cost equals*

$$q \;+\; \sum_{l=2}^{p-1} \big|\, C_{l+1} \cap (C_1 \cup \dots \cup C_{l-1}) \setminus C_l \,\big|.$$

In particular a three-configuration path (C_a, C_b, C_c) costs $q + |(C_a \cap C_c) \setminus C_b|$.

Proof. By the cost identity (lemma 7.1) the cost is $q + R$ with R the re-insertions. A tool is re-inserted at step $l + 1$ iff it enters C_{l+1} (hence lies outside C_l) having been present before, i.e. it lies in $C_1 \cup \dots \cup C_{l-1}$. Summing over steps gives the formula. □

Proposition 7.8 (Exact heuristic cost at $K^* = 3$)**.** *Let $K^* = 3$. Then*

$$H \;=\; q \;+\; \min\, \min\bigl(x_{ab},\, x_{bc},\, x_{ca}\bigr), \qquad x_{uv} := |(C_u \cap C_v) \setminus C_w| \quad (\{u, v, w\} = \{a, b, c\}),$$

the outer minimum over feasible 3-groupings and configuration choices $C_a, C_b, C_c \subseteq U$. Consequently gap $= \min\{\cdot\} - R$*, and* gap $= 0$ *iff some minimum-cardinality grouping admits configurations with* $\min(x_{ab}, x_{bc}, x_{ca}) \le R$.

Proof. At $K^* = 3$ the three configurations of any feasible 3-grouping are distinct (two equal configurations would merge their groups into a feasible 2-grouping) and cover U. By lemma 7.7 the ordering with C_w in the middle costs $q + x_{uv}$; minimising over the three orderings gives $q + \min(x_{ab}, x_{bc}, x_{ca})$, and H minimises this over groupings and configuration choices. □

Grouping-selection reading. At $K^* = 3$ the criterion is now explicit: choose the grouping (and configurations) minimising the smallest pairwise overlap outside the third configuration; the target to meet is R.

Lemma 7.9 (Tight-coverage walks)**.** *If $Z^*_{\mathrm{SSP}} = q$, some optimal schedule's walk of distinct configurations has $p \le q+1$ configurations, each serving at least one job; hence $K^* \le p \le q+1$, and if $p = K^*$ then* gap $= 0$*.*

Proof. $R = 0$, so each of the $p-1$ transitions inserts at least one tool and every insertion is first-time; there are q of those, so $p - 1 \le q$. A configuration serving no job can be excised: d satisfies the triangle inequality $d(A, C) \le d(A, B) + d(B, C)$, and the cost cannot fall below q (theorem 2.2), so the shortened walk is still optimal with $R = 0$. Assigning every job to a configuration serving it exhibits a feasible p-grouping, whence $p \ge K^*$. If $p = K^*$ that grouping is minimum-cardinality and $H \le Z^*_{\mathrm{SSP}}$. □

Theorem 7.10 ($K^* = 3$: proved regimes of conjecture 7.6)**.** *Let $K^* = 3$. Then* gap $\le$ *$K^* - 2 = 1$ holds in each of the following regimes:*

(i) $R \ge q - 1$; indeed gap $\le q - R$ *unconditionally.*
(ii) $Z^*_{\mathrm{SSP}} = q$ and $q \le 2$; indeed gap $= 0$*.*
(iii) $Z^*_{\mathrm{SSP}} = q = 3$, provided the walk of lemma 7.9 has $p = 3$ configurations (then gap $= 0$*) or, at $p = 4$, two of its four groups have combined tool requirement of size at most b.*

Hence at $K^* = 3$ the conjecture remains open only for $Z^*_{\mathrm{SSP}} = q \ge 4$; for $1 \le R \le q - 2$; and, at $Z^*_{\mathrm{SSP}} = q = 3$, for walks whose four groups are pairwise unmergeable.

Proof. (i) By lemma 7.2, $H \le 2\min(b, q) \le 2q$, so gap $= H - q - R \le q - R$, which is ≤ 1 when $R \ge q - 1$.

(ii) By lemma 7.9, $p \le q + 1 \le 3 = K^* \le p$, so $p = K^*$ and the gap vanishes.

(iii) $p \le 4$; if $p = 3$ the gap vanishes by lemma 7.9. Let $p = 4$ with walk $D_1, \ldots, D_4$, groups G_l (the jobs served at D_l) and requirements $U_l \subseteq D_l$. Since $R = 0$ and $q = 3$, each transition inserts exactly one fresh tool: $D_{l+1} \setminus D_l = \{x_{l+1}\}$ and $D_l \setminus D_{l+1} = \{e_{l+1}\}$; and $R = 0$ forces *persistence*: $t \in D_i \cap D_j$ $(i < j)$ implies $t \in D_l$ for all $i \le l \le j$. Suppose $|U_i \cup U_j| \le b$ for some $i < j$; merge G_i and G_j (keeping the other two groups with their walk configurations) — a feasible 3-grouping, hence of minimum cardinality. In each case a configuration $C \supseteq U_i \cup U_j$ and an ordering give wrap ≤ 1 by lemma 7.7:

- $(1, 2)$: $C \subseteq D_1 \cup D_2$ (size b+1, so C exists); order (C, D_3, D_4): wrap $\subseteq (D_1 \cap D_4 \setminus D_3) \cup (D_2 \cap D_4 \setminus D_3) \subseteq \varnothing \cup \{e_3\}$.
- $(3, 4)$: $C \subseteq D_3 \cup D_4$; order (D_1, D_2, C): wrap $\subseteq (D_1 \cap D_3 \setminus D_2) \cup (D_1 \cap D_4 \setminus D_2) = \varnothing$ by persistence.
- $(2, 3)$: $W := U_2 \cup U_3 \cup (D_1 \cap D_4) \subseteq D_2 \cup \{x_3\}$ (by persistence $D_1 \cap D_4 \subseteq D_2 \cap D_3$), so $|W| \le b$+1; take $C \supseteq U_2 \cup U_3$ inside W, dropping (if $|W| = b$+1) one element outside $U_2 \cup U_3$; order (D_1, C, D_4): wrap $= |(D_1 \cap D_4) \setminus C| \le 1$.
- $(1, 3)$: $C \subseteq D_1 \cup D_3$; order (C, D_2, D_4): wrap $\subseteq \varnothing \cup (D_3 \cap D_4 \setminus D_2) \subseteq \{x_3\}$.
- $(2, 4)$: $C \subseteq D_2 \cup D_4$; order (D_1, D_3, C): wrap $\subseteq (D_1 \cap D_2 \setminus D_3) \cup \varnothing \subseteq \{e_3\}$.
- $(1, 4)$: $C \subseteq D_1 \cup D_4$; order (C, D_2, D_3): wrap $\subseteq \varnothing \cup (D_4 \cap D_3 \setminus D_2) \subseteq \{x_3\}$.

In every case $\gamma \le q + 1$, so $H \le Z^*_{\mathrm{SSP}} + 1$. □

The 6-ring realises case (iii) with a mergeable pair ($U_2 \cup U_3 = \{3, 4, 5\}$) and wrap exactly 1: the theorem is tight there.

Lemma 7.11 (Overlap counting). *Any three configurations $C_a, C_b, C_c \subseteq U$ of size b with $C_a \cup C_b \cup C_c = U$ satisfy*

$$\min\big(x_{ab},\, x_{bc},\, x_{ca}\big) \;\le\; \left\lfloor \tfrac{2b-q}{3} \right\rfloor, \qquad x_{uv} = |(C_u \cap C_v) \setminus C_w|.$$

Exhaustive enumeration attains equality at every tested (b, q).

Proof. Count tools by the number of configurations containing them: with a tools in exactly one configuration, $\sum x := x_{ab} + x_{bc} + x_{ca}$ in exactly two, and c_3 in all three, the tool count gives $a + \sum x + c_3 = b + q$ and the degree sum gives $a + 2\sum x + 3c_3 = 3b$; subtracting, $\sum x = 2b - q - 2c_3 \le 2b - q$. The minimum is at most the mean $\sum x/3$, and it is an integer. □

Theorem 7.12 (Quantitative $K^* = 3$ bound). *If $K^* = 3$ then*

$$\text{gap} \;\le\; \max\Big(0,\; \min\big(q,\; \lfloor \tfrac{2b-q}{3} \rfloor\big) - R\Big),$$

tight at the 6-ring and at the $b = 4$ extremal witness. Consequently:

- *(a) conjecture 7.6 holds at $K^* = 3$ for* every *instance with $b \le 4$ (for $b = 4$, $q = 2$, $R = 0$ combine with theorem 7.10(ii));*
- *(b) at $K^* = 3$ the bound* gap ≤ 1 *can fail only in the corner $2b \ge q + 3R + 6$ with $R \le q - 2$ — necessarily $b \ge 5$ — and there only outside the mergeable case of theorem 7.10(iii); it* does *fail there (refutation box below);*
- *(c) at $b = 3$ the former assumption (a) of proposition 7.16 is discharged:* gap ≤ 1 *is a theorem for $K^* = 3$, $b = 3$.*

Proof. The configurations of any feasible 3-grouping (one exists) cover U, so proposition 7.8 with lemma 7.11 gives $H \le q + \lfloor (2b - q)/3 \rfloor$, while lemma 7.2 gives $H \le 2\min(b, q) \le q + q$; subtract $Z^*_{\text{SSP}} = q + R$ and use gap ≥ 0. For (a) at $b = 3$: $\lfloor (6 - q)/3 \rfloor \le 1$ for all $q \ge 1$. At $b = 4$: the bound is ≤ 1 for $q \ge 3$; for $q \le 2$, either $R \ge 1$ (bound $2 - R \le 1$) or $Z^*_{\text{SSP}} = q \le 2 = K^* - 1$ and theorem 7.10(ii) gives gap 0; $q \le 1$ is the zero-gap class. (b) reads off the display: bound ≥ 2 needs $\lfloor (2b - q)/3 \rfloor \ge R + 2$ and $q \ge R + 2$. (c) is (a) restricted to $b = 3$. □

A first general-K^* bound

Lemma 7.13 (Runs identity). *Let $C_1, \ldots, C_p \subseteq U$ be distinct size-b configurations covering U, in an order σ. The free-initial path cost equals $q + \sum_{t \in U}(\text{runs}_\sigma(t) - 1)$, where $\text{runs}_\sigma(t)$ is the number of maximal blocks of consecutive configurations containing t.*

Proof. By the cost identity the cost is $q + R$; every block start after t's first block is a re-insertion of t, and these are all the re-insertions. □

Theorem 7.14 (Random-ordering bound, all K^*)**.** *For every instance with $K^* \geq 2$,*

$$\text{gap} \;\leq\; \max\Big(0,\; \Big\lfloor \frac{q\,\big(b(K^*-1)-q\big)}{b+q} \Big\rfloor \;-\; R\Big).$$

At $K^*=3$ this and theorem 7.12 are incomparable (each is tighter on part of the range); both are tight at the 6*-ring (*$\lfloor 3\cdot 3/6 \rfloor = 1$*).*

Proof. Fix a minimum-cardinality grouping; its configurations $C_1,\dots,C_{K^*} \subseteq U$ are pairwise distinct (else merge two groups) and cover U. Let d_t be the number of configurations containing t, so $\sum_t d_t = K^*b$ over $|U| = b+q$ tools. For a uniformly random ordering, $\mathbb{E}[\text{runs}(t)] = d_t/K^* + d_t(K^*-d_t)/K^*$, hence $\mathbb{E}[\text{runs}(t)-1] = (d_t-1)(K^*-d_t)/K^*$. The map $d \mapsto (d-1)(K^*-d)$ is concave, so by Jensen

$$\sum_t \frac{(d_t-1)(K^*-d_t)}{K^*} \;\leq\; \frac{b+q}{K^*}\Big(\frac{K^*b}{b+q}-1\Big)\Big(K^*-\frac{K^*b}{b+q}\Big) \;=\; \frac{q\,\big(b(K^*-1)-q\big)}{b+q}.$$

Some ordering achieves at most the mean; the excess-runs total is an integer; apply lemma 7.13 and subtract $Z^*_{\text{SSP}} = q+R$. ☐

The bound grows linearly in K^*, so it does not by itself approach the conjectured K^*-2; but it is unconditional, valid for every K^*, and strictly better than the transition cap $(K^*-1)\min(b,q) - q - R$ whenever $q < b(K^*-1)/2$-ish territory—e.g. for the $b=5$, $K^*=4$ extremal witness it gives $\lfloor 44/9 \rfloor - 0 = 4$ against the cap's 8. Derandomising the ordering (greedy block-preserving orders) and replacing Jensen by the true extremal degree profile are the two places where it can be tightened.

Proposition 7.15 (Consequences of conjecture 7.6)**.** *If conjecture 7.6 holds, then for every instance and every b:*

$$\frac{H}{Z^*_{\text{SSP}}} \;\leq\; 1+\frac{K^*-2}{\max(K^*-1,\,q)} \;\leq\; 2-\frac{1}{K^*-1} \;<\; 2,$$

i.e. the JGP + GSP heuristic is a $(2-\varepsilon)$-approximation universally *in b. Moreover all known extremal witnesses have $q = K^*$ and* gap $= K^*-2$*, where the first bound evaluates to* $2-2/K^*$ *— reproducing exactly the observed extremal ratios* $4/3$ *(K^*=3) and* $3/2$ *(K^*=4).*

Proof. Divide $H \leq Z^*_{\text{SSP}} + K^* - 2$ by $Z^*_{\text{SSP}} \geq \max(K^*-1, q)$; the second inequality takes $q \geq 1$. At $q = K^*$: $1 + (K^*-2)/K^* = 2 - 2/K^*$. ☐

[**Void as of 2026-07-02:** the hypothesis conjecture 7.6 is refuted — see the box in section 7.4. Kept for the record.]

Proposition 7.16 (A route to the 4/3 conjecture)**.** *Assume only that every positive-gap $b=3$ instance has $K^*=3$ and $Z^*_{\text{SSP}} \geq 3$ (as holds throughout the* 10691*-instance edge family, where positive gap forces $K^*=3$, $|U|=6$, $Z^*_{\text{SSP}}=3$; open in general). Then conjecture 5.4 holds: $H/Z^*_{\text{SSP}} \leq 4/3$ for $b=3$. (The earlier draft also assumed conjecture 7.6; that assumption is discharged at $b=3$ by theorem 7.12(c).)*

Proof. Zero-gap instances have ratio 1. For a positive-gap instance, (a) and (b) give gap $\le K^* - 2 = 1$ and $H/Z^*_{\mathrm{SSP}} = 1 + \mathrm{gap}/Z^*_{\mathrm{SSP}} \le 1 + 1/3 = 4/3$. □

Open problems raised by this section.

- *The exact extremal law* (replaces "prove conjecture 7.6", which is now refuted). At $K^* = 3$ every known extremal instance attains $\min(q, \lfloor (2b-q)/3 \rfloor) - R$ (theorem 7.12); decide whether that is the true worst case. At $K^* \ge 4$ the best bound is theorem 7.14; the lever is extending lemma 7.11 from triples to K^*-tuples with the order-dependent wrap sum of lemma 7.7 (the counting identity $\sum x = 2b - q - 2c_3$ has a K^*-analogue; the min-over-orderings does not). Directed search on the proved-region boundaries found both refuting witnesses; continue exactly there.
- *Achievability of ratio* 2. proposition 7.15 caps the ratio at $2 - 1/(K^* - 1)$; approaching 2 needs $K^* \to \infty$ with $q \approx K^* - 1$ and near-maximal gap *simultaneously*. The g-copy construction fails: its tools decouple, Z^*_{SSP} grows linearly per copy, and the ratio *decreases* to 7/6. A family with a *shared* bridge structure across unboundedly many groups (or a proof that none exists, yielding a bound strictly below 2) would settle $\sup_b \sup_I H/Z^*_{\mathrm{SSP}}$.
- *The per-b extremal curve.* Known suprema: 4/3 (b=3), $\ge 4/3$ (b=4), $\ge 3/2$ (b=5), all of the form $2 - 2/K^*$ at witnesses with $q = K^*$. Determine $\sup_I H/Z^*_{\mathrm{SSP}}$ as a function of b; in particular whether $q = K^* - 1$ extremal instances (which would give the larger value $2 - 1/(K^* - 1)$) exist for any b.

Conflict-Graph Structure: What It Does and Does Not Predict

Define the *pairwise conflict graph* G_{conf} on the jobs, with an edge (j, j') whenever $|T_j \cup T_{j'}| > b$. Every feasible grouping is a proper colouring of G_{conf} (a group contains no conflicting pair), so

$$\chi(G_{\mathrm{conf}}) \;\le\; K^*.$$

Remark 8.1 (K^* is not the chromatic number)**.** The inequality can be strict. For the star $\{t_1, t_2\}, \{t_1, t_3\}, \{t_1, t_4\}$ ($b = 3$) we have $\chi(G_{\mathrm{conf}}) = 1$ (no conflicting pair) but $K^* = 2$ (the joint union has size 4). Equality $\chi = K^*$ holds when the tool sets enjoy a Helly-type property under which pairwise compatibility implies joint feasibility — e.g. the ring instances below. The identification "$K^* = \chi(G_{\mathrm{conf}})$" is therefore valid only under such a hypothesis, not in general.

Remark 8.2 (Conflict-graph structure of the rings, corrected)**.** For the 6-ring, two jobs conflict iff they are tool-disjoint, i.e. non-adjacent on the ring. The resulting graph is 3-regular with 9 edges: it is the *triangular prism* $\overline{C_6}$ (the complement of the 6-cycle), **not** $2K_3$. (Earlier notes stated $2K_3$; that is incorrect — e.g. $j_1 = \{t_1, t_2\}$ and $j_4 = \{t_4, t_5\}$ are disjoint and hence *do* conflict.) The prism is perfect, being the complement of an even cycle. For the 5-ring the conflict graph is the imperfect odd cycle C_5 ($\chi = 3$, $\chi_f = 5/2$).

Remark 8.3 (Neither perfectness nor the JGP LP gap predicts the SSP gap)**.** The 6-ring has a *perfect* conflict graph (the prism) yet gap = 1; the 5-ring has an *imperfect* conflict graph (C_5) yet gap = 0 (here $K^* = 3$, $Z^*_{\mathrm{SSP}} = 2$, $H = 2$, both lower bounds met). Likewise the JGP LP relaxation value $\chi_f(G_{\mathrm{conf}})$ does not track the SSP gap: for the 5-ring $\chi_f = 5/2 < 3 = K^*$ (a fractional JGP gap) but the SSP gap is 0, whereas for the 6-ring $\chi_f = 3 = K^*$ (no fractional JGP gap) but the SSP gap is 1. Thus the SSP gap is governed by tool-overlap structure at group boundaries (the bridge mechanism of definition 5.2), not by global chromatic or perfectness properties. A positive structural characterisation remains open. References on perfection and clutters: [Chudnovsky et al.(2006)Chudnovsky, Robertson, Seymour, and Thomas, Cornuéjols(2001)].

Connections to Solution Methods

The quantities here interface with the exact methods developed in the other Parts but are logically separate from them:

- the GSP is a path-GTSP, solvable by GTSP/TSP machinery (Noon–Bean, LKH, branch-and-cut) as surveyed in [Pop et al.(2024)Pop, Cosma, Sabo, and Pop Sitar]; this is the engine of phase 2;
- the non-compact configuration ILP and its column generation — including why a Dantzig–Wolfe master over job sequences degenerates, and the set-covering CG for the JGP — belong to Part II (`02_noncompact_formulation.tex`);
- the MTZ alternative to subtour constraints is Part IV (`04_mtz_formulation.tex`);
- the branch-and-Benders exact solver (master TSP + KTNS subproblem) is Part VIII (`08_branch_and_benders.tex`), which reaches the same Z^*_{SSP} from the sequence side.

The bounds of section 2 (especially $Z^*_{\mathrm{SSP}} \geq \max(K^* - 1, |U| - b)$) are the natural primal bounds to seed any of these solvers.

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